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Faculty of Information Technology and Artificial Intelligence

Movie Rental Data Warehouse Design

From OLTP Sakila Database to Dimensional Model and ETL Design

Course	Data Warehouse
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Date	16 May 2026

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1. Introduction

This report presents a high-level data warehouse design for a movie rental business using the Sakila OLTP database as the source system. The goal is to reorganize operational data into a structure that is easier to use for reporting, analysis, and business decision-making.

The Sakila OLTP database is designed for daily operations. It records individual activities such as customer registration, film inventory, rentals, returns, payments, staff work, and store information. This design is good for fast transaction processing, but it is not ideal for analytical questions such as revenue trends, top films, customer activity, store performance, and late returns.

The proposed data warehouse uses dimensional modeling. It organizes data into fact tables and dimension tables. Fact tables store measurable business events, while dimension tables store descriptive information used to filter and group the facts. This makes the data easier to understand and easier to query for management reports.

Area	OLTP Sakila System	Data Warehouse Design
Main purpose	Supports daily operations and transactions.	Supports reporting, analysis, and decision-making.
Data structure	Many normalized tables with many joins.	Fact tables connected to descriptive dimensions.
Typical users	Operational users and applications.	Managers, analysts, and decision-makers.
Typical questions	Is this rental stored? Was this payment recorded?	Which films generate the highest revenue? Which stores perform best?
Query style	Short transaction queries.	Aggregation queries over time, film, customer, store, staff, and location.

2. Business Questions

Before designing the data warehouse, the business questions must be clear. The following questions represent the main analysis that the movie rental business needs. They cover rental activity, revenue, customer behavior, film popularity, staff performance, store performance, location analysis, and time trends.

No.	Business Question	Why It Is Important
1	Which films are rented most frequently?	Helps management understand film popularity and demand.
2	Which films generate the highest revenue?	Shows which films contribute most to income, not only rental count.
3	Which film categories are most popular?	Supports film purchasing, promotion, and inventory planning.
4	How does revenue change by month, quarter, and year?	Helps identify seasonality and business growth trends.
5	How does rental activity change over time?	Shows busy periods and helps plan staffing and inventory.
6	Which stores generate the highest number of rentals?	Compares store activity and operational performance.
7	Which stores generate the highest revenue?	Compares stores using financial performance.
8	Which customers rent the most films?	Identifies active customers for loyalty programs.
9	Which customers generate the highest revenue?	Identifies high-value customers for marketing and retention.
10	Which staff members process the most rentals or payments?	Helps evaluate staff workload and performance.

11	Which cities or countries generate the most customer activity?	Supports location-based marketing and store planning.
12	What is the average rental duration by film or category?	Helps understand customer viewing behavior and inventory return speed.
13	Which films are returned late most often?	Helps manage late return risk and improve rental policies.
14	How does store performance differ by location?	Combines location and store analysis for better regional decisions.
15	What are the most active customer locations?	Supports targeted advertising and customer segmentation.

The selected questions are important because they are not focused on single transactions. They require aggregation across many rentals, payments, customers, films, stores, and dates. This is exactly the type of work a data warehouse is designed to support.

3. Dimensional Model Design

The dimensional model is built around the main business events in the movie rental process. In this design, the main measurable events are rental transactions and payment transactions. These events become fact tables. The descriptive data around these events becomes dimension tables.

3.1 Selected Business Processes

Business Process	Reason for Selection	Fact Table
Rental transaction	A rental is the main activity of the business. It records which customer rented which film, from which store, handled by which staff member, and when it was returned.	fact_rental
Payment transaction	A payment records revenue. It is needed to analyze total revenue by film, customer, store, staff, and time.	fact_payment

3.2 Fact Tables

Fact tables store numeric measures and foreign keys to dimensions. Each fact table has a clearly defined grain. The grain is important because it tells us exactly what one row in the fact table represents.

Fact Table	Business Process	Grain	Main Measures	Related Dimensions
fact_rental	Rental transaction	One row per rental transaction.	rental_count, rental_duration_days, late_return_flag, late_days	dim_date, dim_customer, dim_film, dim_store, dim_staff
fact_payment	Payment transaction	One row per payment transaction.	payment_amount, payment_count	dim_date, dim_customer, dim_film, dim_store, dim_staff

Fact table details

Column Type	fact_rental	fact_payment
Primary key	rental_fact_key	payment_fact_key
Degenerate key	rental_id from the source system	payment_id from the source system
Date foreign keys	rental_date_key and return_date_key	payment_date_key
Other foreign keys	customer_key, film_key, store_key, staff_key	customer_key, film_key, store_key, staff_key
Measures	rental_count, rental_duration_days, late_return_flag, late_days	payment_amount, payment_count
Main analysis supported	Rental volume, rental duration, late returns, film demand, customer activity, staff/store activity	Revenue by time, film, customer, staff, and store

3.3 Dimension Tables

Dimension tables provide descriptive context for facts. They allow users to group and filter the measures by date, customer, film, store, staff, and location. Surrogate keys are used as primary keys in the warehouse dimensions. The original OLTP IDs are kept as business keys for traceability.

Dimension	Primary Key	Main Attributes	Source OLTP Tables	Purpose
dim_date	date_key	full_date, day, month, month_name, quarter, year, day_of_week	rental, payment	Supports analysis by day, month, quarter, and year.
dim_location	location_key	address, address2, district, city, country, postal_code, phone	address, city, country	Supports city, country, district, and address-based analysis.
dim_customer	customer_key	customer_id, customer_name, email, active, create_date, location_key	customer, address, city, country	Supports customer activity, customer revenue, and location analysis.
dim_film	film_key	film_id, title, description, release_year, rental_duration, rental_rate, length, rating, replacement_cost, language, categories, actors	film, language, film_category, category, film_actor, actor	Supports film popularity, category analysis, actor-related analysis, and revenue by film.
dim_store	store_key	store_id, manager_staff_id, manager_name, location_key	store, staff, address, city, country	Supports store performance and store location analysis.
dim_staff	staff_key	staff_id, staff_name, email, active, username, store_key, location_key	staff, store, address, city, country	Supports staff workload and payment/rental processing analysis.

3.4 Shared Dimensions

Shared dimensions are dimensions used by more than one fact table. They make reporting consistent across different business processes. For example, the same customer dimension is used for both rental analysis and payment analysis. This allows the business to compare rental behavior and revenue for the same customers.

Shared Dimension	Used By	Reason
dim_date	fact_rental and fact_payment	Supports time-based rental and revenue trends.
dim_customer	fact_rental and fact_payment	Allows customer activity and customer revenue analysis.
dim_film	fact_rental and fact_payment	Allows film popularity and film revenue analysis.
dim_store	fact_rental and fact_payment	Allows store activity and store revenue analysis.
dim_staff	fact_rental and fact_payment	Allows staff workload and payment processing analysis.

3.5 Schema Type and Design Decisions

This design uses a simplified star schema. The two fact tables are placed at the center of the model, and the dimension tables are connected directly to the facts. This structure is simple, clear, and suitable for analytical reporting.

The model is called a simplified star schema because some descriptive data is denormalized into a single dimension. For example, category, actor, and language information are stored inside dim_film. This reduces the number of joins needed for common reports. It also makes the model easier to understand for this assignment.

In a larger production warehouse, film categories and actors could be modeled with separate dimensions and bridge tables because films can have many actors and can belong to categories through many-to-many relationships. For this high-level assignment, storing all film categories and actors as comma-separated descriptive attributes in dim_film is a reasonable simplification.

Design Decision	Reason
Use two fact tables	Rentals and payments are different business events and have different measures.
Use surrogate keys	Surrogate keys protect the warehouse from changes in operational IDs and simplify joins.
Use shared dimensions	Shared dimensions allow consistent reporting across rental and revenue analysis.
Denormalize category, actor, and language into dim_film	This simplifies the schema and makes common film reports easier.
Use an unknown date row with date_key 19000101	This handles rentals that have not been returned yet.

Keep rental_id and payment_id as degenerate dimensions

These source transaction IDs are useful for tracing facts back to the OLTP system.

4. Dimensional Model Diagram

The following diagram shows the proposed dimensional model. The model contains two fact tables and six dimension tables. The fact tables store measurable events, while dimensions provide descriptive context.

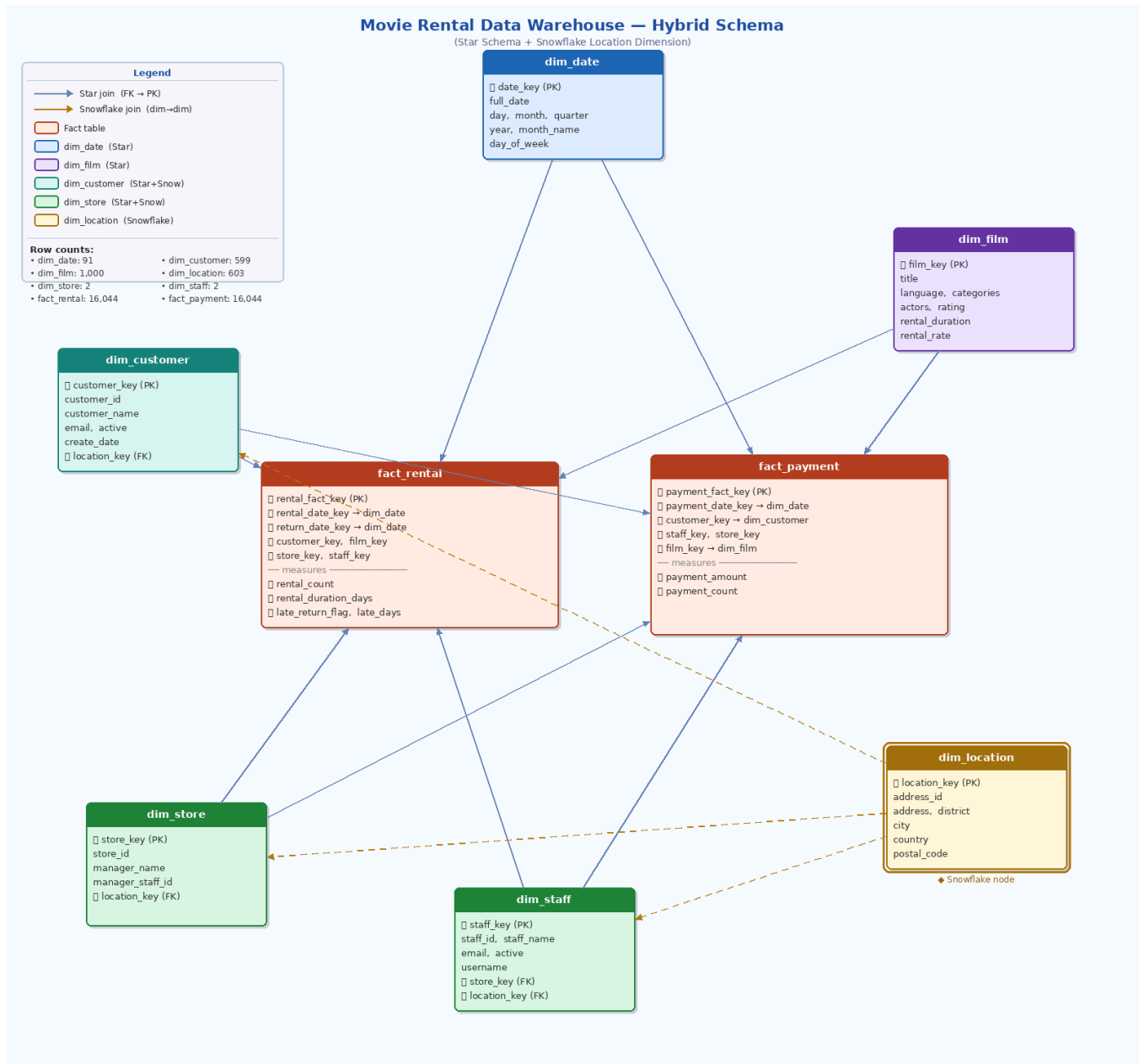


Figure 1. Movie Rental Data Warehouse dimensional model diagram.

5. ETL Design

The ETL process moves data from the Sakila OLTP database into the movie_rental_dw data warehouse. The process first extracts source tables, then transforms them into clean dimensions and facts, and finally loads the final tables into MySQL.

5.1 Extract

The extract step reads the required operational tables from the Sakila source database. Only tables needed for analysis and dimensional modeling are extracted.

Source Table	Why It Is Needed
rental	Provides rental transactions, rental dates, return dates, customer IDs, inventory IDs, and staff IDs.
payment	Provides payment transactions, payment amount, payment date, customer ID, staff ID, and rental ID.
customer	Provides customer names, emails, active status, create date, store ID, and address ID.
film	Provides film title, description, release year, rental duration, rental rate, length, rating, replacement cost, and language ID.
inventory	Connects rentals to films and stores through inventory ID.
store	Provides store ID, manager staff ID, and store address.
staff	Provides staff names, email, active status, username, store ID, and address ID.
address	Provides address, district, postal code, phone, city ID, and location details.
city	Connects addresses to city names and countries.
country	Provides country names for location analysis.
category	Provides film category names.
film_category	Connects films to categories.
language	Provides film language names.
actor	Provides actor names.
film_actor	Connects films to actors.

5.2 Transform

The transform step changes normalized OLTP data into analytical warehouse tables. This step is the most important part of the ETL process because it prepares the data for reporting.

Transformation	Explanation
Duplicate checking	Each source table is checked for duplicate rows. Duplicate rows are removed if found.
Missing-value checking	Source null values are reviewed. Known acceptable nulls are handled during transformation.
Date key creation	Rental, return, and payment dates are converted into integer date keys in YYYYMMDD format.
Unknown date handling	Unreturned rentals use date_key 19000101 so that every fact row has a valid return_date_key.
Surrogate key generation	Each dimension receives a warehouse key such as customer_key, film_key, store_key, and staff_key.
Customer name creation	Customer first and last names are combined into customer_name.
Staff and manager name creation	Staff first and last names are combined into staff_name and manager_name.
Location creation	Address, city, and country are joined to create dim_location.
Film enrichment	Film is joined with language, category, and actor data to create a descriptive dim_film.
Many-to-many handling	Film categories and actors are grouped into comma-separated attributes inside dim_film.
Rental duration calculation	The number of days between rental_date and return_date is calculated. Unreturned rentals use -1.
Late return calculation	Actual rental duration is compared with expected film rental_duration to calculate late_days and late_return_flag.
Payment film key creation	Payment is joined to rental and inventory so revenue can be analyzed by film.
Measure creation	rental_count and payment_count are created as additive count measures.

5.3 Load

The load step writes the final warehouse tables into the target MySQL database named movie_rental_dw. Dimension tables are loaded before fact tables because fact tables reference dimension surrogate keys.

1. Load dim_date first, including the unknown date row.
2. Load dim_location because customer, store, and staff dimensions use it.
3. Load dim_customer, dim_film, dim_store, and dim_staff.
4. Load fact_rental and fact_payment after all dimensions are available.
5. Add primary keys after loading because pandas to_sql recreates the tables.

6. Run row-count validation to confirm that important source transactions were not lost.

In the implementation notebook, the load process uses pandas to_sql with if_exists='replace'. This is acceptable for an assignment implementation because it rebuilds the warehouse tables from the current source data. For a production warehouse, the process could be changed to incremental loading.

6. Data Quality Considerations

Data quality checks are required to make sure the warehouse is reliable. The notebook includes checks before and after loading. These checks reduce the risk of broken joins, missing keys, duplicate keys, and incorrect measures.

Data Quality Rule	Reason	Action in This Design
Check duplicate source rows	Duplicate rows can produce incorrect counts and revenue totals.	Each source table is checked with duplicated(). Duplicates are removed if found.
Review missing values	Null values can break joins or calculations.	Known acceptable nulls are documented. Important fields are handled during transformation.
Handle missing return_date	Some rentals are not returned yet.	Use return_date_key 19000101 and rental_duration_days = -1.
Check missing foreign keys in fact_rental	Facts must connect correctly to dimensions.	Assert that customer_key, film_key, store_key, and staff_key are not missing.
Check missing foreign keys in fact_payment	Revenue analysis must connect to customer, staff, store, and film.	Assert that customer_key, staff_key, store_key, and film_key are not missing.
Check negative payment amounts	Negative revenue would be invalid for normal payment records.	Assert that payment_amount is not negative.
Check duplicate dimension surrogate keys	Duplicate primary keys would make joins unreliable.	Assert that all dimension surrogate keys are unique.
Check duplicate business keys	One customer_id or film_id should not map to multiple current rows in this simplified design.	Assert that customer_id and film_id are not duplicated in their dimensions.
Validate row counts after loading	The ETL should not lose source transactions.	Compare dim_film, fact_rental, and fact_payment counts with source tables.

7. Conclusion

This report proposed a complete high-level data warehouse design for the Sakila movie rental OLTP system. The design converts the operational schema into a simplified star schema that is better for business analysis and decision-making.

The model focuses on two important business processes: rental transactions and payment transactions. These processes are represented by fact_rental and fact_payment. The fact tables are connected to shared dimensions for date, customer, film, store, staff, and location. This structure supports analysis of film popularity, revenue, rental trends, customer behavior, store performance, staff performance, location activity, and late returns.

The ETL design extracts the required Sakila tables, transforms them into analytical dimensions and facts, and loads them into the movie_rental_dw database. Data quality checks are included to validate missing values, duplicate keys, missing foreign keys, negative payments, and row-count consistency.

Appendix A. Example Analytical Queries Supported by the Warehouse

The data warehouse can answer the following analytical questions using simple joins between fact and dimension tables.

Question	Main Tables Used	Example Measure
Top 10 most rented films	fact_rental, dim_film	COUNT(rental_fact_key)
Revenue by month	fact_payment, dim_date	SUM(payment_amount)

Store performance by rentals	fact_rental, dim_store	COUNT(rental_fact_key)
Top films by revenue	fact_payment, dim_film	SUM(payment_amount)
Films with most late returns	fact_rental, dim_film	COUNT late rental rows and AVG(late_days)
Top customers by revenue	fact_payment, dim_customer	SUM(payment_amount)
Customer activity by country	fact_rental, dim_customer, dim_location	COUNT(rental_fact_key)